

Earth-Venus-Earth Modem

Operator's Guide

Rev A - DRAFT

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DSES

September 2026

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Earth-Venus-Earth Modem — Operator's Guide

Document	DSES EVE Modem Operator's Guide
Revision	Rev A — DRAFT
Date	2026-09-12
Prepared by	Rick Hambly, K0GD, Deep Space Exploration Society
Companion	DSES EVE Modem Design Description and ICD (Rev C): the why behind every setting
Where it lives	This text is the application's Help → Operator's guide, and docs/DSES_EVE_Modem_Operators_Guide.pdf

This guide is for the person at the keyboard at Haswell. It says what to connect, what to click, what to watch, and what to do when something is wrong. The design document explains the reasons; this one gives the steps.

1. What the program does

The DSES EVE modem sends an 11-symbol message (K0PRT K0PRT) as one tone at a time out of 4096 possible tones, bounces it off the Moon or Venus, and decides which tones came back. One symbol lasts 164.8 seconds (473 frames of 0.35 s); a whole message takes 30.2 minutes per pass, and several passes are added together before the decision. The program does all of it from one window:

- **Setup:** pick the kind of run, set the frequency, gains, message, and timing; preview the schedule; press Start.
- **Run:** watch the tones arrive, the decisions form, the chunks progress, the key line, the radio and the GPS clock.
- **Report:** read the session report when the run ends, re-decode an earlier one, or tick Compare to put two reports side by side.

The settings are remembered between runs, so the desktop icon opens the program ready for the last test. A run can be repeated, changed, and repeated again without restarting the program.

2. Before you start: the hardware checklist

Item	What to check
B210	On a USB 3 port of the PC (a direct rear port, not the powered extension cable). It appears in Device Manager under “USRPs”. After a power event that leaves it unrecognized, unplug USB and the DC barrel for 15 s (a true cold start).
GPS reference clock (Leo Bodnar)	OUT1 to the B210 REF IN (10 MHz). OUT2 to the B210 PPS IN (with output 2 disabled the OUT2 connector carries the 1 PPS; the program sets this). USB to the PC. GPS antenna with a clear sky view. Both LEDs steady after a few minutes = locked.
Key line	B210 header J504, pin 1 (GPIO_0) and a ground pin (9 or 10) , 3.3 V logic, through the isolated driver to the sequencer. On the clone with the case connector: pin “1” and a “G”. Never straight to the sequencer input.
Transmit	B210 TX/RX A to the driver’s input pad. The B210 gives at most +8 dBm; the driver decides the TX gain setting (Setup → TX gain).
Receive	LNA output (through the bandpass filter if fitted) to B210 RX2 A . Nothing on RX2 for the loopback bench.
Sequencer	LNA sequencer and LNA DC control in place; the amplifier’s own interlocks armed; the 2 kW load only for the thermal test.
PC clock	The PC needs internet NTP (the second number of the PPS-edge time set comes from it). Windows: Settings → Time → Sync now.
Pointing	The dish tracks the target under System 1 (RA/Dec, J2000). The Run tab shows the target’s azimuth and elevation from the ephemeris for a cross-check.
Archive folder	A folder with room for the run: about 24 MB per 30-minute pass at the modem rate.

3. Starting the program

Double-click the **DSES EVE Modem** desktop icon (created once with `install-shortcut.ps1`). The window opens on the Setup tab with the last settings. The status bar shows where the settings file is. If the program does not start, the log is in `%LOCALAPPDATA%\DSES\EVE_Modem\app.log`.

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Setup

Run

Report

Run mode

☒ Software simulation (no radio)

☐ Bench loopback (one R210, no antenna)

☐ EME — Earth-Moon-Earth

☐ EVE — Earth-Venus-Earth

Waveform and message

Variant

A

Dial frequency

1296.0000 MHz

Message

KOPRT KOPRT

Repeat count

1

Pilot

☒ pilot frames before each pass

Symbol length

☐ full length

test length:

4 frames

☐ pilot

2 frames

4 frames per symbol = 1.4 s; one message pass = 0.3 min (TEST length; no on-air significance)

Radio and reference (R210)

R210 serial

first R2xx found

TX gain

0.00 dB

RX gain

30.00 dB

Clock source

external

Time

☐ host-timed (no FPS available; epoch from the PC's NTP clock)

GPS clock

☒ program and check the Leo Bodnar GPS clock first (OUT1 10 MHz, OUT2 off = 1 FPS)

LO offset

-300.00 kHz

Archive folder

C:\Users\rick\AppData\Local\Temp\eve_app_archive

Browse...

Software simulation

C/N0

25.00 dB-Hz

Range

0.1 km

Noise seed

1

Schedule preview

SIM-20260912-221031: sim, bistatic_tx, f_dial 1296.0000 MHz
frames 44 of 44 wanted = 1 complete messages; 8 chunks;
chunk 0: frames 0- 5 (6) tx 22:10:31-22:10:33
chunk 1: frames 6- 11 (6) tx 22:10:33-22:10:36
chunk 2: frames 12- 17 (6) tx 22:10:36-22:10:38
chunk 3: frames 18- 23 (6) tx 22:10:39-22:10:41
chunk 4: frames 24- 29 (6) tx 22:10:41-22:10:43
chunk 5: frames 30- 35 (6) tx 22:10:44-22:10:46
chunk 6: frames 36- 41 (6) tx 22:10:46-22:10:49
chunk 7: frames 42- 43 (2) tx 22:10:49-22:10:50

Preview schedule

START

ABORT

Save settings now

22:10:19 SIM-20260912-221031: sim, bistatic_tx, f_dial 1296.0000 MHz
frames 44 of 44 wanted = 1 complete messages; 8 chunks;
chunk 0: frames 0- 5 (6) tx 22:10:31-22:10:33
chunk 1: frames 6- 11 (6) tx 22:10:33-22:10:36
chunk 2: frames 12- 17 (6) tx 22:10:36-22:10:38
chunk 3: frames 18- 23 (6) tx 22:10:39-22:10:41
chunk 4: frames 24- 29 (6) tx 22:10:41-22:10:43
chunk 5: frames 30- 35 (6) tx 22:10:44-22:10:46
chunk 6: frames 36- 41 (6) tx 22:10:46-22:10:49
chunk 7: frames 42- 43 (2) tx 22:10:49-22:10:50
duration 0.3 min (0h 00m); symbol 4 frames = 1.4 s; message 1
22:10:33 session finished: aborted=False; chunks keyed 1
22:10:33 offline decode (decision of record) ...
22:10:33 SIM-20260912-221031_00.eve.iq: 98307 samples (2.10 MB)
22:10:33 SIM-20260912-221031_01.eve.iq: 98307 samples (2.10 MB)
22:10:33 SIM-20260912-221031_02.eve.iq: 98306 samples (2.10 MB)
22:10:33 SIM-20260912-221031_03.eve.iq: 98307 samples (2.10 MB)
22:10:34 SIM-20260912-221031_04.eve.iq: 98307 samples (2.10 MB)
22:10:34 SIM-20260912-221031_05.eve.iq: 98306 samples (2.10 MB)
22:10:34 SIM-20260912-221031_06.eve.iq: 98307 samples (2.10 MB)
22:10:34 SIM-20260912-221031_07.eve.iq: 32769 samples (0.65 MB)
22:10:34 OFFLINE DECODE: OK 'KOPRT KOPRT' symbols [1203, 1203, 1203, 1203, 1203, 1203, 1203, 1203]
22:10:34 report written: C:\Users\rick\AppData\Local\Temp\eve_app_archive\221031_00.eve.iq

idle

settings: C:\Users\rick\AppData\Local\Temp\eve_app_cfg\DSSEVE_Modem.ini

Setup tab

4. The Setup tab

Every control has a tooltip: hover over it or its label. The essentials:

4.1 Run mode

Mode	When to use it
Software simulation	No radio. Proves the decoder and the display at a chosen signal-to-noise. Run it first on any new PC.
Bench loopback	One B210, nothing on the antenna ports. Transmits at low gain, receives its own internal leakage. Proves the radio, the reference, the archive, and the decoder before anything goes on the air.
Interop with a partner station	Compatibility test with ORI's own hardware and software, on the bench through a cable and attenuator or across the room: transmit only (their receiver decodes us) or receive only (their generator transmits, we archive and decode). No Doppler, no round trip, no amplifier limits. See section 7.1.
EME	Moon bounce. Ephemeris from JPL Horizons, Doppler pre-compensated, transmit 2.4 s then listen for the 2.5 s echo, repeat. The rehearsal before Venus.
EVE	Venus bounce. 240 s transmit chunks against the 272 s round trip. A full message is 30.2 minutes per pass; plan five passes.

4.2 Waveform and message

- **Variant** A is the ORI air interface (2.87 Hz bins, 473 frames per symbol). Use A whenever another station must decode us. B is a DSES-only variant.
- **Dial frequency:** 1299.5 MHz for Venus 2026 with the 23 cm package; 1296 on the bench. The comb sits 25 to 48.5 kHz above the dial.
- **Message:** K0PRT K0PRT, the club call. Up to 11 characters.
- **Repeat count:** passes of the message. The receiver adds them; five for Venus.
- **Symbol length: full length** (473 frames, 164.8 s per symbol) is the air interface. The **test length** (6 frames, 2.1 s) makes a whole message pass in 23 s for bench work. The program warns before sending a test-length signal on the air.

4.3 Radio and reference

- **TX gain** 0 dB on the bench; on the air, what the driver's input pad expects.
- **RX gain:** set so the background of the tone strip is visible but not saturated (30 dB on the bench, 40 to 50 dB behind the LNA).
- **Clock source** external (the GPS clock on REF IN and PPS IN). The run refuses to start unless the B210 reports lock.
- **Time:** leave host-timed unticked when the PPS is connected. Tick it only if there is no PPS; the epoch then comes from the PC clock, still inside the tolerance.

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- **GPS clock:** ticked, the program sets the Leo Bodnar clock to the station setting and waits for lock before opening the radio.
- **Archive folder:** where the run's files go.

4.4 Mode settings

The group under the radio settings changes with the mode. For EME and Venus the sky session group adds the start time (now plus a lead, or a UTC time), the ephemeris source (Horizons, astropy fallback), the station mode (monostatic, or transmit-only for a partner receiver), and the Doppler handling. The defaults are the design values; change them only for a reason you can write in the log.

4.5 Preview, Start, Abort

Preview schedule builds the schedule without touching the radio and lists every chunk with its times, frames, round trip, elevation, and Doppler, then the total duration. For a sky session this fetches the ephemeris, which takes a few seconds. Check the elevation and the duration before Start.

START saves the settings, programs the GPS clock, opens the radio, builds the schedule, and switches to the Run tab. The first chunk starts after the lead time (15 s).

ABORT releases the key line at once, stops the streams, closes the archive, and writes the log. Use it for any fault. The same button is on the Run tab.

5. The Run tab: what to watch

Setup

Run

Report

SIM-20260912-221031 · sim · bistatic_tx · 1296.0000 MHz · Variant A · 'K0PRT K0PRT' · repeat 1 · 4 frames/symbol (1.4 s)

TX chunk 2 of 8

UTC 22:10:35

device 22:10:35

Tone strip (frames x candidate bins, newest at top). 4096-ary FSK sends ONE tone per frame: one dot per row is the whole signal

Running decisions (live accumulator; the offline decode is the decision of record)

symbol	expected	decided	margin dB	frames	state
0	1203	1203	16.1	2	✓
1	80	80	16.1	2	✓
2	1317	1317	18.2	4	✓
3	1056	1056	16.2	2	✓
4	1203	1203	14.7	2	✓
5	80	80	16.9	4	✓
6	1317	1317	15.6	2	✓
7	1031	1031	16.2	2	✓
8	289	289	17.7	4	✓
9	3894	3894	15.6	2	✓
10	3168	3168	15.8	2	✓

frames 44 passes [0] symbols right 11/11 BCH ok (0 corrected) CRC ok
text 'K0PRT K0PRT'

Schedule

chunk 1 of 8: done

chunks complete 1/8 session 22:10:31 .. 22:10:50 UTC remaining 0:14

frames wanted 44 received 44

KEY DOWN — TRANSMITTING

Radio and ephemeris

SimRadio: rate 1504706.56 S/s, C/N0 25.0 dB-Hz

GPS clock D65EE73B3D: sat LOCK, PLL LOCK, signal losses 0

sim: az 180.0 el 45.0 RTT 0.00 s Doppler -0.0 Hz rate +0.000 Hz/s

ABORT — release key stop

22:10:19 SIM-20260912-221031: sim, bistatic_tx, f_dial 1296.0000 MHz, variant A, message 'K0PRT K0PRT', repeat 1, pilot on
frames 44 of 44 wanted = 1 complete messages; 8 chunks; 2026-09-12T22:10:31Z .. 2026-09-12T22:10:50Z (0.3 min)
chunk 0: frames 0- 5 (6) tx 22:10:31-22:10:33 rx 22:10:31-22:10:33 rtt 0.0 s el 45.0 dop -0 Hz rate +0.000 Hz/s
chunk 1: frames 6- 11 (6) tx 22:10:33-22:10:36 rx 22:10:33-22:10:36 rtt 0.0 s el 45.0 dop -0 Hz rate +0.000 Hz/s
chunk 2: frames 12- 17 (6) tx 22:10:36-22:10:38 rx 22:10:36-22:10:38 rtt 0.0 s el 45.0 dop -0 Hz rate +0.000 Hz/s
chunk 3: frames 18- 23 (6) tx 22:10:39-22:10:41 rx 22:10:39-22:10:41 rtt 0.0 s el 45.0 dop -0 Hz rate +0.000 Hz/s
chunk 4: frames 24- 29 (6) tx 22:10:41-22:10:43 rx 22:10:41-22:10:43 rtt 0.0 s el 45.0 dop -0 Hz rate +0.000 Hz/s
chunk 5: frames 30- 35 (6) tx 22:10:44-22:10:46 rx 22:10:44-22:10:46 rtt 0.0 s el 45.0 dop -0 Hz rate +0.000 Hz/s
chunk 6: frames 36- 41 (6) tx 22:10:46-22:10:49 rx 22:10:46-22:10:49 rtt 0.0 s el 45.0 dop -0 Hz rate +0.000 Hz/s
chunk 7: frames 42- 43 (2) tx 22:10:49-22:10:50 rx 22:10:49-22:10:50 rtt 0.0 s el 45.0 dop -0 Hz rate +0.000 Hz/s
duration 0.3 min (0h 00m); symbol 4 frames = 1.4 s; message 0.3 min per pass
22:10:19 session SIM-20260912-221031: 8 chunks, first at 2026-09-12T22:10:31Z, flowgraph at 2026-09-12T22:10:30Z (in 10.8 s)

Run tab

12-Sep-26

8

Panel	What a healthy run looks like
Phase (top right)	armed → TX chunk n of N → listening for chunk n → ... → finished.
Clocks	UTC (PC) and device time agree to milliseconds when the time was set on a PPS.
Tone strip	One bright dot per row (one tone per frame) that steps to a new column at each symbol boundary; a steady column during the pilot. The background is receiver noise. A blank strip means no signal, wrong RX gain, or the receive window is closed (between chunks). At Venus strength single frames show nothing: that is expected.
Current symbol	The sum over the frames of the symbol in progress. The gold dashed line is the tone we sent; the red line is the leader. On the bench they coincide from the first frame; on Venus the peak climbs out of the noise as frames add up.
Running decisions	One row per symbol: expected, decided, margin in dB, frames summed. Green rows agree with what we sent. The line under the table shows the message the live accumulators would decode now.
Schedule	Chunk in progress, chunks complete, time remaining, frames wanted vs received.
Key lamp	Red while the key line is asserted. It rises 200 ms before the RF and drops 100 ms after.
Radio and ephemeris	LOCKED must be shown on the air. LO offset ok in both directions. GPS clock: sat LOCK, PLL LOCK. Target azimuth, elevation, round trip, Doppler, and rate.
Log	Every step the program took, keying events, radio messages, decode results.

The live decisions are a view for the operator; the decision of record is the offline decode of the archived windows that runs when the session ends and appears in the report.

6. The Report tab

SetupRunReport

Session reports in the archive folder (newest first)

SIM-20260912-221031 2026-09-12 16:10
SIM-20260912-220440 2026-09-12 16:04

Refresh

Open in PDF viewer

Open archive folder

Re-decode this session

Zoom 100 %

idle

settings: C:/U

DSES Earth-Venus-Earth modem — session report

SIM-20260912-221031

sim · bistatic_tx · 1296.0000 MHz · Variant A · message 'K0PRT K0PRT' · repeat 1

DECODED

offline decode (decision of record): text 'K0PRT K0PRT' BCH correcte

started 2026-09-12T22:10:19Z finished 2026-09-12T22:10:33Z

schedule 2026-09-12T22:10:31Z .. 2026-09-12T22:10:50Z chunks 8 frames wante
symbol 4 frames = 1.4 s (TEST length); message 0.3 min per pass; bins 2.87 Hz;
samples in 1,026,539 written 720,916 windows filed 8 gap events 0 gap sampl
live decode (operator view): ok 'K0PRT K0PRT' frames 44 passes [0]

Radio: SimRadio: rate 1504706.56 S/s, C/N0 25.0 dB-Hz
options: PA limits off, TX pre-compensation off, RX Doppler removal off, archiv

Symbols (combined over passes)

symbol	expected	decided	margin dB	per pass
0	1203	1203	16.1	1203
1	80	80	16.1	80
2	1317	1317	18.2	1317
3	1056	1056	16.2	1056
4	1203	1203	14.7	1203
5	80	80	16.9	80
6	1317	1317	15.6	1317
7	1031	1031	16.2	1031
8	289	289	17.7	289
9	3894	3894	15.6	3894
10	3168	3168	15.8	3168

Report tab

When a run ends the program decodes the archive, writes `<session>_report.pdf` beside the session log, and shows it here: the verdict, the decisions with margins per pass, the chunk timeline with key events, the per-window synchronization (pilot offset, tracker residual, sample gaps), and the key-event list.

- **Open in PDF viewer** for printing or sending.
- **Open archive folder** to reach the raw files.
- **Re-decode this session** runs the offline decode again on the archived windows and rebuilds the report (after a software fix, or to try without the pilot).
- Earlier reports in the same folder are listed; click one to view it. The teal bar over the pages names the report shown.
- **Compare** opens a second pane with its own chooser, so two runs sit side by side (Zoom 50 % fits both on a laptop screen).

7. The test sequence at the site

1. **Software simulation** at C/N0 20 dB-Hz, test-length symbols: proves the program on the site PC. One minute.
2. **Bench loopback**, test length, TX gain 0, GPS clock ticked, clock source external: proves the B210, the reference lock, the PPS time set, the archive, the decode. About two minutes. Both passes should decode at 20 dB or more of margin.
3. **Bench loopback, full length, one pass** (34 minutes): proves the long streams with no underruns and the amplifier-limit chunking at real length. Run it once per PC.
4. **EME**, test length first, then full length, with the dish on the Moon: the echo path, the Doppler pre-compensation, the sequencer, the LNA, and the gains. Expect the echo in the receive windows 2.5 s after each chunk.
5. **Venus**: full length, five passes, start time from the session plan.

After every run, read the report before deciding the next step. Keep the archive folder; the raw windows can be decoded again later.

7.1 The interop test with ORI

Both ends agree on four things beforehand: Variant A with full-length symbols, the dial frequency, the message (K0PRT K0PRT), and the UTC start of the transmission. In the program choose **Interop with a partner station**, tick **full length**, and set the start either as “now + lead” (tell the partner the time the preview shows) or as the agreed UTC time.

- **Transmit only:** we send the message; the partner’s receiver decodes it. Their result is the test result. Our report shows what left the radio.
- **Receive only:** the partner transmits from their generator, which has no pilot: **untick Pilot**. We send no RF and key nothing. After the run the decoder searches for their actual start within the **start search range** (30 frames = ± 10 s around the agreed time; widen it if their start is looser) and the report says by how many frames they were off and whether the message decoded.

Cable and attenuator: the B210 puts out up to +8 dBm, and its receiver wants no more than about –20 dBm at RX2, so 30 to 40 dB of attenuation between two radios on the bench.

8. When something is wrong

Symptom	What it means and what to do
“GPS clock not locked; refusing to start”	The Leo Bodnar clock has no satellite or PLL lock. Check the antenna’s sky view and the LEDs; wait a few minutes after power-up; after reprogramming the PLL drops for a few seconds and relocks within 20 s.
“reference not locked (external)”	No 10 MHz at REF IN, or the clock is not locked. Check the OUT1 cable and the clock’s LEDs.
“USRP time did not take on the PPS edge”	No PPS at PPS IN. Check OUT2 to PPS IN (output 2 must be disabled: the program does that). As a fallback tick host-timed.
“PPS verify False” in the log	Same as above: no second PPS edge was seen.
A storm of ‘L’ or LATE_COMMAND in the log at the start	The radio was not armed in time. Increase the lead time (15 s or more) and close other programs.
‘O’ or overflows in the log	The PC dropped receive samples. One short overflow at start-up is known and padded; repeated ones mean the PC is too busy: close other programs, use a direct USB 3 port. The archive records every gap.
Blank tone strip during a receive window	No signal or wrong RX gain. On the bench check TX gain 0 and nothing on the antenna ports; on the air check the LNA and the sequencer.
Decisions wrong on the bench	Reference unlocked (frequency off), or the LO offset fell back (the radio line says so). Do not go on the air until the bench decodes.

Symptom	What it means and what to do
“a chunk of N frames cannot hold the pilot frames plus data”	The chunk is too short for the pilot. Variant B frames are 0.667 s, so the 2.4 s bench chunk holds three frames and two are pilot: use a chunk of 4 s or more for Variant B (the message says the minimum).
“chunk would hold no frames”	The chunk settings leave no room between the round trip and the guard. Restore the mode's defaults.
Horizons unreachable	The ephemeris source falls back to astropy with the local DE440s. The Doppler differs by up to 10 Hz at 13 cm; acceptable, but note it in the log.
The program closes by itself	A native library crashed. Two files under %LOCALAPPDATA%\DSES\EVE_Modem tell the story: app.log (everything the program printed) and fault.log (the traceback of a hard crash). Send both. If the B210 was left streaming by a program that was killed, power-cycle it (USB and DC off for 15 s) before the next try.
The program will not start from the icon	Read %LOCALAPPDATA%\DSES\EVE_Modem\app.log. The project environment must exist in .conda next to the program.

9. Files a run produces

In the archive folder, all named by the session id (mode and UTC start):

- <session>.json — the schedule (the contract with any partner station).
- <session>_<target>_haswell.csv — the ephemeris table used (sky sessions).
- <session>_NN.eve.iq + .json — one receive window per chunk, complex float samples at the modem rate with a sidecar (start time, first frame, gaps).
- <session>_session.json — the session log (times, keying, radio, live decode).
- <session>_report.pdf — the report.

10. Command-line equivalents

The window drives the same code as the tools in tools/: eve_session.py plan | run | sim, eve_bench.py, eve_decode.py, eve_txcw.py (one comb tone into the lab counter), and python -m eve.gpsdo status | config | preflight for the GPS clock. Use them for scripted tests and for the field log; the design document section 5.1 lists their options.